Ovule number, pollen competition, and directional reproductive isolation in plants

A theory note, reciprocal-cross audit, and analysis plan

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Abstract

Here, I develop a model for how post-pollination prezygotic reproductive isolation can arise in flowering plants. The mechanism is the number of effective pollen grains competing per ovule in an ovary. When that ratio is high, pollen grains compete for limited fertilisation opportunities. Selection can then favour pollen traits that win fertilisations and maternal traits that filter pollen more strictly. Reproductive isolation arises as a by-product when the evolved pollen and pistil traits of one lineage no longer work well with those of another. The main empirical prediction is directional. Reciprocal crosses should fail more strongly when the high-competition lineage is the maternal parent. I then audit the current data structure and show that the strongest immediate test is an asymmetric reciprocal-cross test, analysed with family-level partial pooling and sensitivity analyses for ovule-number confidence, genetic distance, mating system, measurement type, and phylogenetic non-independence. The current data support a useful first analysis, but not a fully saturated model.

Contents

1 Purpose of this note

I want a model that does three jobs. First, it must state the biological mechanism in a testable form. Second, it must show how ovule number, pollen load, mating system, and pollen-pistil mismatch enter the same set of equations. Third, it must lead to predictions that can be separated from older explanations, especially unilateral incompatibility.

The starting idea is simple. An ovary has a finite number of ovules. A stigma receives pollen. If the number of effective pollen grains is smaller than or similar to the number of ovules, most compatible pollen grains can fertilise. Competition among pollen grains is weak. If the number of effective pollen grains is much larger than the number of ovules, some pollen grains must lose. That losing margin creates selection.

Working metaphor

Think of ovules as seats in a small room. Pollen grains are applicants for those seats. When there are as many applicants as seats, the filter can be loose. When there are many more applicants than seats, the filter can be strict, and even small differences among applicants matter.

The model is not that few ovules directly cause reproductive isolation. The model is that few ovules can move a lineage into a high-competition reproductive state. In that state, selection on pollen performance and maternal filtering becomes stronger. Reproductive isolation can then appear as a passenger of that selection. Ovule number can be thought of as a modulator of the strength of sexual selection in plants.

Conclusion

The causal claim is not

$$\text{few ovules} \Rightarrow \text{reproductive isolation.}$$

The causal claim is

$$\begin{aligned}\text{few ovules} &\Rightarrow \text{higher pollen competition}\\ &\Rightarrow \text{stronger pollen and pistil selection}\\ &\Rightarrow \text{directional prezygotic isolation.}\end{aligned}$$

The middle terms are the mechanism.

Note that this model does not predict the evolution of directional postzygotic isolation, a consideration that we explain below.

2 Biological background

Speciation studies often treat reproductive isolation as something that increases with time. This view works well for many postzygotic barriers. Hybrid sterility and hybrid inviability can increase as substitutions accumulate between lineages. In plants, however, post-pollination prezygotic barriers are not only historical residues. They also depend on the current reproductive arena.

After pollen lands on a stigma, it must hydrate, germinate, enter maternal tissues, grow through the style, respond to guidance cues, reach ovules, and release sperm cells. These steps occur inside maternal tissue. They allow selection on pollen traits and on maternal filtering traits (pers. comm. Marcy Uyenoyama, 2007). The pollen tube pathway in angiosperms has therefore been viewed as a setting for pollen competition and selection (Lora et al., 2016; Lankinen and Karlsson Green, 2015). Reviews of post-pollination barriers also emphasise that pollen-pistil interactions can reject heterospecific pollen and contribute to prezygotic reproductive isolation (Wang and Filatov, 2023).

The important point for this model is that the pistil is not a pipe. It is a selective environment. That environment may become more selective when pollen abundance exceeds ovule abundance.

Conclusion

The pistil is part of the selective environment. This matters because a maternal trait can create a directional barrier. Genetic distance is symmetric. A cross direction is not.

3 Notation

Symbol	Meaning
O_i	Number of ovules per ovary in lineage i.
A_i	Pollen production per relevant unit. The unit must match the biological scale of pollen delivery.
ψ_i	Effective deposition fraction. It combines pollen transfer, germination, and timing.
$P_{\text{eff},i}$	Effective pollen load. This is the pollen that can compete for the same ovules.
K_i	Effective pollen grains per ovule, $K_i = P_{\text{eff},i}/O_i$.
F_i	Strictness of the maternal filter in lineage i.
D_{ij}	Pollen-pistil mismatch between lineages i and j.
G_{ij}	Genetic distance or divergence measure between lineages i and j.
C_{ii}	Conspecific reproductive success for maternal lineage i.
$H_{i\leftarrow j}$	Heterospecific reproductive success for cross mother i, father j.
$R_{i\leftarrow j}$	Directional reproductive isolation index for cross mother i, father j.
A_{ij}	Reciprocal asymmetry, $A_{ij} = R_{i\leftarrow j} - R_{j\leftarrow i}$.
t_i	Outcrossing rate of lineage i.
s_i	Self-incompatibility status of lineage i, where $s_i = 1$ means SI and $s_i = 0$ means SC.

4 The competition ratio

4.1 Definition

The central quantity is

$$K_i = \frac{P_{\text{eff},i}}{O_i}. \tag{1}$$

Here $P_{\text{eff},i}$ is not total pollen production. It is not even necessarily the total pollen counted on the stigma. It is the pollen that can plausibly compete for the same ovules. A pollen grain contributes to $P_{\text{eff},i}$ only if it arrives in the relevant time window, remains viable, germinates, and enters the competitive pathway.

Definition

K_i is the number of effective pollen grains per ovule. It is a competition index, not a pollen count.

A useful decomposition is

$$P_{\text{eff},i} = \psi_i A_i, \tag{2}$$

where A_i is pollen production and ψ_i is the effective deposition fraction. We can write

$$\psi_i = \delta_i \gamma_i \tau_i. \tag{3}$$

Here δ_i is the fraction of produced pollen that reaches a focal stigma, γ_i is the fraction of deposited pollen that germinates or enters the pollen-tube pool, and τ_i measures temporal overlap among competing pollen grains.

Substituting (2) into (1) gives

$$K_i = \frac{\psi_i A_i}{O_i}. \tag{4}$$

Caution

A fixed pollen-deposition fraction, such as $\psi = 0.02$, is useful for simulation. It should not be treated as a biological constant. It is a calibration value.

4.2 Why the ratio matters

When $K_i \leq 1$, the effective pollen load is no larger than the ovule number. Pollen may still fail for physiological reasons, but there is little numerical pressure for pollen grains to exclude one another.

When $K_i > 1$, the effective pollen load exceeds the number of ovules. Only a subset of pollen grains can fertilise. This creates the opportunity for selection among pollen grains.

One simple index of that opportunity is

$$\Omega_i = \mathbb{E}\left[\frac{(P_{if} - O_i)_+}{P_{if}}\right], \tag{5}$$

where P_{if} is the effective pollen load on flower f and

$$(x)_+ = \begin{cases} x, & x > 0, \\ 0, & x \leq 0. \end{cases} \tag{6}$$

The numerator $P_{if} - O_i$ is the number of pollen grains that cannot win an ovule when pollen exceeds ovules. The denominator P_{if} turns that number into a fraction.

Conclusion

K says whether competition is possible on average. Ω says how much pollen is excluded at the flower level. Both quantities should be high in the strongest version of the model.

5 Sublinear pollen-load scaling

The model needs one scaling condition. Effective pollen load should increase with ovule number more slowly than one-to-one. Write

$$P_{\text{eff},i} = \psi c O_i^{\alpha}, \tag{7}$$

where $c > 0$ and α is the scaling exponent.

Substitute (7) into the definition of K_i:

$$K_i = \frac{P_{\text{eff},i}}{O_i} \tag{8}$$

$$= \frac{\psi c O_i^{\alpha}}{O_i}. \tag{9}$$

Now use the exponent rule

$$\frac{O_i^{\alpha}}{O_i} = \frac{O_i^{\alpha}}{O_i^{1}} = O_i^{\alpha-1}. \tag{10}$$

Therefore

$$K_i = \psi c O_i^{\alpha-1}. \tag{11}$$

Rule used

The exponent rule used above is

$$\frac{x^a}{x^b} = x^{a-b}.$$

Here $a = \alpha$ and $b = 1$.

Take logs of both sides of (11):

$$\log K_i = \log(\psi c O_i^{\alpha-1}) \tag{12}$$

$$= \log \psi + \log c + \log(O_i^{\alpha-1}) \tag{13}$$

$$= \log(\psi c) + (\alpha - 1) \log O_i. \tag{14}$$

Differentiate (14) with respect to $\log O_i$:

$$\frac{\partial \log K_i}{\partial \log O_i} = \alpha - 1. \tag{15}$$

If $\alpha < 1$, then $\alpha - 1 < 0$. Thus

$$\frac{\partial \log K_i}{\partial \log O_i} < 0. \tag{16}$$

Conclusion

If effective pollen load scales sublinearly with ovule number, then the pollen-per-ovule ratio declines as ovule number rises. Few-ovule lineages have the stronger competition regime.

5.1 The competition threshold

The threshold between weak and strong numerical competition is $K_i = 1$. Use (11):

$$1 = \psi c O_i^{\alpha-1}. \tag{17}$$

Move ψc to the left-hand side:

$$\frac{1}{\psi c} = O_i^{\alpha-1}. \tag{18}$$

Because $\alpha - 1 = -(1-\alpha)$, we can write

$$O_i^{\alpha-1} = O_i^{-(1-\alpha)} = \frac{1}{O_i^{1-\alpha}}. \tag{19}$$

Thus

$$\frac{1}{\psi c} = \frac{1}{O_i^{1-\alpha}}. \tag{20}$$

Invert both sides:

$$\psi c = O_i^{1-\alpha}. \tag{21}$$

Raise both sides to the power $1/(1-\alpha)$:

$$O^* = (\psi c)^{1/(1-\alpha)}. \tag{22}$$

Conclusion

There is no universal ovule threshold. The threshold depends on pollen production, pollen transfer, germination, timing, and the scaling exponent α.

6 Selection among pollen grains

6.1 Saved fraction

Assume a flower receives P effective pollen grains and has O ovules. If $P > O$, then only O pollen grains can win. The fraction of pollen grains that win is

$$q = \frac{O}{P}. \tag{23}$$

Since $K = P/O$, we have $P = KO$. Substitute this into the equation for q:

$$q = \frac{O}{KO} \tag{24}$$
$$= \frac{1}{K}. \tag{25}$$

Conclusion

When $K = 2$, half the pollen grains can win. When $K = 10$, one tenth can win. As K rises, the winners come from a smaller and more extreme part of the pollen-performance distribution.

6.2 Truncation selection

Let z be pollen competitive performance. This could represent pollen-tube growth rate, success in maternal tissue, response to ovule guidance, or a weighted combination of such traits. For the first derivation, assume

$$z \sim N(\bar{z}, \sigma_z^2). \tag{26}$$

We can standardise z as

$$x = \frac{z - \bar{z}}{\sigma_z}, \tag{27}$$

so that

$$x \sim N(0, 1). \tag{28}$$

If only the top fraction q of pollen grains succeeds, then the cutoff on the standard normal scale is

$$t = \Phi^{-1}(1 - q), \tag{29}$$

where Φ^{-1} is the standard normal quantile function.

The mean of a standard normal variable above cutoff t is

$$\mathbb{E}[x \mid x > t] = \frac{\phi(t)}{1 - \Phi(t)}. \tag{30}$$

But $1 - \Phi(t) = q$. Therefore

$$\mathbb{E}[x \mid x > t] = \frac{\phi(t)}{q}. \tag{31}$$

Return to the original scale. Since $z = \bar{z} + \sigma_z x$,

$$\mathbb{E}[z \mid z > z_t] = \bar{z} + \sigma_z \frac{\phi(t)}{q}. \tag{32}$$

The selection differential is the mean of successful pollen minus the original mean:

$$S_z = \mathbb{E}[z \mid z > z_t] - \bar{z} \tag{33}$$
$$= \sigma_z \frac{\phi(t)}{q}. \tag{34}$$

Now use $q = 1/K$ from (25). Then

$$t = \Phi^{-1}\left(1 - \frac{1}{K}\right), \tag{35}$$

and

$$S_z(K) = \sigma_z K \phi\left[\Phi^{-1}\left(1 - \frac{1}{K}\right)\right] \quad \text{for } K > 1. \tag{36}$$

Set $S_z(K) = 0$ for $K \leq 1$. The full expression is

$$S_z(K) = \begin{cases} 0, & K \leq 1, \\ \sigma_z K \phi\left[\Phi^{-1}\left(1 - \frac{1}{K}\right)\right], & K > 1. \end{cases} \tag{37}$$

Rule used

The calculation uses the mean of a truncated normal distribution. For a standard normal variable x, the mean above cutoff t is

$$\mathbb{E}[x \mid x > t] = \frac{\phi(t)}{1 - \Phi(t)}.$$

The denominator is the saved fraction.

6.3 From selection differential to allele-frequency change

The selection gradient on pollen performance is

$$\beta_z(K) = \frac{S_z(K)}{\sigma_z^2}. \tag{38}$$

This says how much relative success changes per unit of pollen performance.

Let allele ℓ increase pollen performance by a_ℓ. A first-order approximation to its male-function selection coefficient is

$$s_\ell^{(m)}(K) \approx a_\ell \beta_z(K). \tag{39}$$

If only part of the pollen-phase advantage is expressed in whole-plant allele-frequency change, write

$$s_\ell^{\text{eff}}(K) = \kappa a_\ell \beta_z(K), \tag{40}$$

where $0 < \kappa \leq 1$.

For an allele at frequency p_ℓ, the standard weak-selection recursion is

$$\Delta p_\ell \approx p_\ell(1 - p_\ell) s_\ell^{\text{eff}}(K). \tag{41}$$

Substitute (40):

$$\Delta p_\ell \approx p_\ell(1 - p_\ell)\kappa a_\ell \beta_z(K). \tag{42}$$

Now substitute $K = \psi c O^{\alpha-1}$:

$$\Delta p_\ell \approx p_\ell(1 - p_\ell)\kappa a_\ell \beta_z\left(\psi c O^{\alpha-1}\right). \tag{43}$$

Conclusion

If $\alpha < 1$, then lower ovule number increases K. Higher K increases the truncation-selection differential. Thus few-ovule lineages should experience stronger selection on pollen-performance alleles, all else equal.

7 Maternal filtering

Selection among pollen grains can change male function. Directional reproductive isolation also needs a maternal component. Let $F_i \geq 0$ be the strictness of the maternal filter in lineage i.

A strict filter has two effects. It can improve offspring quality by rejecting unsuitable pollen. It can also reduce seed set when pollen is scarce. The balance between these effects depends on K.

7.1 A simple fitness model for the filter

Let the accepted fraction of pollen decline with filter strictness as e^{-F}. Then the accepted pollen-per-ovule ratio is

$$Ke^{-F}. \tag{44}$$

Use a saturating seed-set function:

$$S(K, F) = 1 - e^{-Ke^{-F}}. \tag{45}$$

This function is near zero when accepted pollen is scarce. It approaches one when accepted pollen is abundant.

Let the offspring-quality benefit of filtering be

$$Q(F) = 1 + b(1 - e^{-\lambda F}), \tag{46}$$

where $b > 0$ is the maximum quality benefit and $\lambda > 0$ controls how fast the benefit rises.

Let the direct cost of strictness be

$$C(F) = e^{-c_F F^2/2}, \tag{47}$$

where $c_F > 0$.

Maternal fitness is

$$W_f(F, K) = S(K, F)Q(F)C(F). \tag{48}$$

Take logs because products become sums:

$$\log W_f(F, K) = \log S(K, F) + \log Q(F) - \frac{c_F F^2}{2}. \tag{49}$$

Rule used

Maximising W is equivalent to maximising $\log W$ when $W > 0$, because the log function is monotonic. We use logs because derivatives of sums are easier than derivatives of products.

7.2 The first-order condition

The optimum filter $F^*(K)$ solves

$$\frac{\partial \log W_f(F, K)}{\partial F} = 0. \tag{50}$$

We now compute each term.

First, define

$$x = Ke^{-F}. \tag{51}$$

Then $S(K, F) = 1 - e^{-x}$. The derivative of x with respect to F is

$$\frac{\partial x}{\partial F} = -Ke^{-F} = -x. \tag{52}$$

The derivative of S with respect to F is

$$\begin{align}
\frac{\partial S}{\partial F} &= \frac{\partial}{\partial F}(1 - e^{-x}) \tag{53}\\
&= e^{-x}\frac{\partial x}{\partial F} \tag{54}\\
&= -xe^{-x}. \tag{55}
\end{align}$$

Therefore

$$\frac{\partial \log S}{\partial F} = \frac{1}{S}\frac{\partial S}{\partial F} = -\frac{xe^{-x}}{1 - e^{-x}}. \tag{56}$$

Second, for $Q(F) = 1 + b(1 - e^{-\lambda F})$,

$$\frac{\partial Q}{\partial F} = b\lambda e^{-\lambda F}. \tag{57}$$

Thus

$$\frac{\partial \log Q}{\partial F} = \frac{b\lambda e^{-\lambda F}}{1 + b(1 - e^{-\lambda F})}. \tag{58}$$

Third,

$$\frac{\partial}{\partial F}\left(-\frac{c_F F^2}{2}\right) = -c_F F. \tag{59}$$

Combine (56), (58), and (59). The optimum satisfies

$$-\frac{xe^{-x}}{1 - e^{-x}} + \frac{b\lambda e^{-\lambda F}}{1 + b(1 - e^{-\lambda F})} - c_F F = 0, \quad x = Ke^{-F}. \tag{60}$$

The first term is the seed-set cost of filtering. It is negative. The second term is the offspring-quality benefit of filtering. It is positive. The third term is the direct cost of filter strictness. It is negative.

As K rises, $x = Ke^{-F}$ rises at any fixed F. The magnitude of the seed-set cost term

$$\frac{xe^{-x}}{1 - e^{-x}} = \frac{x}{e^x - 1} \tag{61}$$

declines as x increases. This means pollen surplus makes filtering cheaper. The optimum filter therefore shifts upward over the parameter range where filtering gives a quality benefit.

Conclusion

High K does not make a strict pistil inevitable. It makes a strict pistil cheaper. If filtering improves offspring quality or avoids poor pollen, selection can then favour a higher F (pers. comm. John Willis and Marcy Uyenoyama, 2007).

For the rest of the model we use the qualitative result

$$\frac{dF^*(K)}{dK} > 0. \tag{62}$$

8 Mating system as a modifier of competition

Mating system changes what K means. A selfing plant can place many pollen grains on its own stigma. That may produce a high total pollen-per-ovule ratio. But if the pollen comes from one genotype, the opportunity for sexual selection among paternal genotypes is limited.

We therefore distinguish total pollen competition from genetically diverse pollen competition.

8.1 Partitioning the pollen pool

Write the effective pollen load as

$$P_{\text{eff},i} = P_{s,i} + P_{x,i} + P_{h,i}, \tag{63}$$

where $P_{s,i}$ is self pollen, $P_{x,i}$ is conspecific outcross pollen, and $P_{h,i}$ is heterospecific pollen.

The total competition ratio is

$$K_{\text{tot},i} = \frac{P_{s,i} + P_{x,i} + P_{h,i}}{O_i}. \tag{64}$$

The outcross competition ratio is

$$K_{\text{out},i} = \frac{P_{x,i}}{O_i}. \tag{65}$$

The costly-pollen ratio is

$$K_{\text{bad},i} = \frac{P_{s,i} + P_{h,i}}{O_i}. \tag{66}$$

Here "bad" means costly in the context of maternal fitness. Self pollen can be costly if inbreeding depression is high. Heterospecific pollen can be costly if it blocks conspecific fertilisation, creates inviable seeds, or wastes ovules.

Now define pollen donor diversity. Let π_{ig} be the fraction of the effective pollen pool on lineage i contributed by donor genotype g. A simple diversity index is

$$D_i^{\text{pollen}} = 1 - \sum_g \pi_{ig}^2. \tag{67}$$

This is close to zero when most effective pollen comes from one genotype. It is larger when many donors contribute.

Define sexually effective competition as

$$K_{\mathrm{sex},i} = K_{\mathrm{tot},i} D_i^{\mathrm{pollen}}. \tag{68}$$

Conclusion

The model should use K_{sex}, not only K_{tot}, when the question is sexual selection among genetically distinct pollen donors. Total pollen can be high in selfers, while sexually effective competition remains low.

8.2 Expected mating-system regimes

The ovule-gated signal should be strongest in lineages where surplus pollen is also genetically diverse. That points to outcrossers and mixed-maters, not predominant autonomous selfers.

Mating system	Effect on K	Expected ovule-gated signal
Obligate outcrossing or strong self-incompatibility	High donor diversity if pollen is abundant. Self pollen is rejected or ineffective.	Strong, but SI itself can confound maternal rejection.
Mostly outcrossing and self-compatible	High outcross pollen with some self pollen.	Strong if outcross pollen is abundant and donor diversity is high.
Mixed mating with competing selfing	Self and outcross pollen arrive in the same window.	Strong. This is a good regime for cryptic maternal filtering.
Mixed mating with delayed selfing	Outcross pollen has the first chance; selfing provides backup.	Moderate to strong. The timing helps preserve outcross competition.
Predominant autonomous selfing	Total pollen can be high, but donor diversity is low.	Weak or absent for this mechanism. RI may evolve through other routes.

Working metaphor

Mating system is a valve. Ovule number sets the number of seats. Pollen production sets the number of applicants. The mating system controls whether the applicants are genetically distinct or mostly copies of the same donor.

8.3 A maternal-filter model with self and outcross pollen

A maternal filter should be favoured when there is enough acceptable pollen to replace rejected pollen and enough costly pollen to make rejection useful. One way to write this is

$$W_i(F) = \left[1 - \exp\left(-\frac{P_{x,i} + e^{-F_s}P_{s,i} + e^{-F_h}P_{h,i}}{O_i}\right)\right]\left[1 - \delta_i \frac{e^{-F_s}P_{s,i}}{P_{x,i} + e^{-F_s}P_{s,i} + e^{-F_h}P_{h,i}}\right]\exp\left(-\frac{c_F F^2}{2}\right). \tag{69}$$

Here F_s is filtering against self pollen, F_h is filtering against heterospecific pollen, and δ_i is inbreeding depression.

The first bracket is seed-set assurance. The second bracket is offspring-quality protection against selfed progeny. The final term is the cost of filter strictness. The equation can be simplified in simulations, but it states the logic.

Conclusion

Strict maternal filtering is most likely when three conditions hold: outcross pollen is abundant, costly pollen is also present, and the fitness cost of accepting costly pollen is high.

8.4 Regression form with mating system

If K is measured or estimated, the empirical model for directed prezygotic success can include an interaction between maternal competition and outcrossing:

$$\operatorname{logit}\left(p^{\text{pre}}_{i\leftarrow j}\right) = \beta_0 + \beta_K \log K_i + \beta_t t_i + \beta_{Kt} \log K_i\, t_i + \beta_G G_{ij} + \cdots. \tag{70}$$

Here t_i is the outcrossing rate of the maternal lineage. The key coefficient is β_{Kt}.

For asymmetry, use

$$A_{ij} = \beta_0 + \beta_1\left[G_{ij}(\log O_j - \log O_i)\right] + \beta_2 \bar{t}_{ij}\left[G_{ij}(\log O_j - \log O_i)\right] + \epsilon_{ij}, \tag{71}$$

where $\bar{t}_{ij} = (t_i + t_j)/2$. The prediction is that the ovule-gated effect should be stronger when $\bar{t}_{ij}$ is high.

Caution

Do not pool selfers, mixed-maters, and outcrossers without modelling mating system. A weak pooled signal could mean the mechanism is absent. It could also mean the mechanism is present only where genetically diverse pollen competition exists.

9 Prezygotic reproductive isolation

9.1 The reproductive isolation index

For the directed cross mother i, father j, let C_{ii} be the conspecific reproductive success of maternal lineage i. Let $H_{i\leftarrow j}$ be the heterospecific success when lineage i provides the ovules and lineage j provides pollen.

Use the Sobel–Chen index (Sobel and Chen, 2014):

$$R_{i\leftarrow j} = 1 - \frac{2H_{i\leftarrow j}}{H_{i\leftarrow j} + C_{ii}}. \tag{72}$$

Define the relative heterospecific success

$$h_{i\leftarrow j} = \frac{H_{i\leftarrow j}}{C_{ii}}. \tag{73}$$

This compares the heterospecific cross with the mother's own conspecific baseline.

Now derive the simplified form. From (73),

$$H_{i\leftarrow j} = h_{i\leftarrow j} C_{ii}. \tag{74}$$

Substitute this into (72):

$$\begin{aligned}
R_{i\leftarrow j} &= 1 - \frac{2h_{i\leftarrow j} C_{ii}}{h_{i\leftarrow j} C_{ii} + C_{ii}} && (75)\\
&= 1 - \frac{2h_{i\leftarrow j} C_{ii}}{C_{ii}(h_{i\leftarrow j} + 1)} && (76)\\
&= 1 - \frac{2h_{i\leftarrow j}}{1 + h_{i\leftarrow j}} && (77)\\
&= \frac{1 + h_{i\leftarrow j}}{1 + h_{i\leftarrow j}} - \frac{2h_{i\leftarrow j}}{1 + h_{i\leftarrow j}} && (78)\\
&= \frac{1 - h_{i\leftarrow j}}{1 + h_{i\leftarrow j}}. && (79)
\end{aligned}$$

Thus

$$\boxed{R_{i\leftarrow j} = \frac{1 - h_{i\leftarrow j}}{1 + h_{i\leftarrow j}}}. \tag{80}$$

Conclusion

The index depends on heterospecific success relative to the mother's own baseline. This is important. It prevents a low-fecundity mother from appearing isolated simply because she has low success in all crosses.

9.2 Mechanistic content of relative success

The ovule-gated model puts the maternal filter into $h_{i\leftarrow j}$. Let D_{ij} be the mismatch between the pollen-pistil systems of lineages i and j. Start with a symmetric mismatch:

$$D_{ij} = D_{ji}. \tag{81}$$

The directed part is the maternal filter F_i. Write

$$h_{i\leftarrow j} = \exp(-F_i D_{ij}). \tag{82}$$

This equation has three assumptions.

1. Mismatch is a property of the pair: $D_{ij} = D_{ji}$.
2. Conspecific mismatch is zero: $D_{ii} = 0$.
3. Filter strictness is a property of the mother: $F_i = F^*(K_i)$.

If a strict filter also lowers conspecific success, some of its effect will cancel when we normalise by C_{ii}. That case is biologically possible. The model should treat it as a sensitivity analysis.

Substitute (82) into (80):

$$R_{i\leftarrow j} = \frac{1 - e^{-F_i D_{ij}}}{1 + e^{-F_i D_{ij}}}. \tag{83}$$

This expression can be written as a hyperbolic tangent:

$$R_{i\leftarrow j} = \tanh\left(\frac{F_i D_{ij}}{2}\right). \tag{84}$$

Rule used

The identity used above is

$$\tanh(x/2) = \frac{1 - e^{-x}}{1 + e^{-x}}.$$

Here $x = F_i D_{ij}$.

10 Directional asymmetry

10.1 Exact expression

Define reciprocal asymmetry as

$$A_{ij} = R_{i\leftarrow j} - R_{j\leftarrow i}. \tag{85}$$

Using (84),

$$A_{ij} = \tanh\left(\frac{F_i D_{ij}}{2}\right) - \tanh\left(\frac{F_j D_{ij}}{2}\right). \tag{86}$$

This is the exact asymmetry under the model.

10.2 The sign theorem

The function $\tanh(x)$ is strictly increasing. If $D_{ij} > 0$, then

$$F_i > F_j \quad \Rightarrow \quad \frac{F_i D_{ij}}{2} > \frac{F_j D_{ij}}{2}. \tag{87}$$

Because tanh is increasing,

$$\tanh\left(\frac{F_i D_{ij}}{2}\right) > \tanh\left(\frac{F_j D_{ij}}{2}\right). \tag{88}$$

Therefore

$$A_{ij} > 0. \tag{89}$$

The same reasoning gives

$$\boxed{\operatorname{sign}(A_{ij}) = \operatorname{sign}(F_i - F_j)}. \tag{90}$$

Conclusion

Symmetric divergence can build a barrier, but it cannot choose a cross direction. Directional asymmetry requires a directional trait. In this model that trait is the maternal filter.

10.3 Small-mismatch approximation

For small x,

$$\tanh(x) \approx x. \tag{91}$$

In (84), $x = F_i D_{ij}/2$. Therefore

$$R_{i\leftarrow j} \approx \frac{F_i D_{ij}}{2}. \tag{92}$$

Apply this to both reciprocal directions:

$$R_{i\leftarrow j} \approx \frac{F_i D_{ij}}{2}, \tag{93}$$

$$R_{j\leftarrow i} \approx \frac{F_j D_{ij}}{2}. \tag{94}$$

Subtract:

$$A_{ij} = R_{i\leftarrow j} - R_{j\leftarrow i} \tag{95}$$

$$\approx \frac{F_i D_{ij}}{2} - \frac{F_j D_{ij}}{2} \tag{96}$$

$$= \frac{D_{ij}}{2}(F_i - F_j). \tag{97}$$

Thus

$$\boxed{A_{ij} \approx \frac{1}{2} D_{ij}(F_i - F_j)}. \tag{98}$$

Conclusion

The factor $1/2$ comes from the Sobel–Chen index. A simpler $1 - C$ index would not have this factor.

10.4 From filter strictness to ovule number

We now connect $F_i - F_j$ to ovule number. The chain is

$$O_i \to K_i \to F_i. \tag{99}$$

From (11),

$$K_i = \psi c O_i^{\alpha-1}. \tag{100}$$

Using (62), F increases with K. We need the derivative of F with respect to $\log O$.

Use the chain rule:

$$\frac{dF}{d\log O} = \frac{dF}{dK}\frac{dK}{d\log O}. \tag{101}$$

We now compute $dK/d\log O$. Since

$$\log K = \log(\psi c) + (\alpha - 1)\log O, \tag{102}$$

we have

$$\frac{d\log K}{d\log O} = \alpha - 1. \tag{103}$$

For any positive variable K,

$$\frac{dK}{d\log O} = K\frac{d\log K}{d\log O}. \tag{104}$$

Therefore

$$\frac{dK}{d\log O} = K(\alpha - 1). \tag{105}$$

So

$$\frac{dF}{d\log O} = F'(K)K(\alpha - 1). \tag{106}$$

Because $F'(K) > 0$ and $\alpha < 1$, this derivative is negative.

Define the positive sensitivity

$$\beta_F = -\frac{dF}{d\log O} = F'(K)K(1 - \alpha) > 0. \tag{107}$$

Then a first-order approximation gives

$$F_i - F_j \approx \beta_F(\log O_j - \log O_i). \tag{108}$$

Rule used

The sign can be checked with a simple example. If lineage i has fewer ovules than lineage j, then $\log O_i < \log O_j$. Thus $\log O_j - \log O_i > 0$. Under sublinear scaling, the fewer-ovule lineage has higher K and higher F, so $F_i - F_j > 0$.

Let mismatch scale with genetic distance to leading order:

$$D_{ij} \approx \delta_D G_{ij}, \tag{109}$$

where $\delta_D > 0$.

Substitute (108) and (109) into (98):

$$\begin{aligned} A_{ij} &\approx \frac{1}{2}D_{ij}(F_i - F_j) && (110)\\ &\approx \frac{1}{2}(\delta_D G_{ij})\left[\beta_F(\log O_j - \log O_i)\right] && (111)\\ &= \left(\frac{1}{2}\beta_F\delta_D\right) G_{ij}(\log O_j - \log O_i). && (112) \end{aligned}$$

Thus

$$\boxed{A_{ij} \approx \beta_1 G_{ij}(\log O_j - \log O_i), \quad \beta_1 = \frac{1}{2}\beta_F \delta_D > 0.} \tag{113}$$

Conclusion

The model predicts an interaction. Asymmetry needs divergence to build mismatch and an ovule contrast to tilt the maternal filter. Remove either component and asymmetry should approach zero.

10.5 Why asymmetry should peak at intermediate divergence

Use the exact expression

$$A(D) = \tanh\left(\frac{F_i D}{2}\right) - \tanh\left(\frac{F_j D}{2}\right), \tag{114}$$

with $F_i > F_j$.

At zero divergence,

$$A(0) = \tanh(0) - \tanh(0) = 0. \tag{115}$$

At very high divergence,

$$\lim_{D\to\infty} A(D) = 1 - 1 = 0. \tag{116}$$

At intermediate divergence, the two curves separate because the stricter maternal filter reaches high RI earlier. Therefore $A(D)$ rises, peaks, and falls.

The derivative is

$$A'(D) = \frac{F_i}{2}\operatorname{sech}^2\left(\frac{F_i D}{2}\right) - \frac{F_j}{2}\operatorname{sech}^2\left(\frac{F_j D}{2}\right). \tag{117}$$

The peak occurs when

$$F_i \operatorname{sech}^2\left(\frac{F_i D}{2}\right) = F_j \operatorname{sech}^2\left(\frac{F_j D}{2}\right). \tag{118}$$

This equation does not need a closed-form solution for the comparative test. It gives the qualitative warning: asymmetry is a mid-divergence signal.

Working metaphor

Divergence builds a wall. The maternal filter decides which side closes first. When the wall is tiny, both directions pass. When the wall is complete, both directions fail. Direction is most visible in the middle.

11 Unilateral incompatibility as an alternative mechanism

The ovule-gated model is not the only way to obtain directional cross failure. Unilateral incompatibility can also create asymmetry. The classical SI–SC rule states that crosses between self-incompatible and self-compatible species often fail when the self-incompatible species is the maternal parent and succeed more often in the reciprocal direction (Lewis and Crowe, 1958; Wang and Filatov, 2023). Molecular work in tomato has linked some unilateral incompatibility to S-locus related factors (Li and Chetelat, 2015).

To compare mechanisms, write unilateral incompatibility in the same coordinates. Let ρ_i be the rejection capacity of the maternal pistil and π_j be the ability of paternal pollen to overcome that rejection. Define

$$h^{\mathrm{UI}}_{i\leftarrow j} = \exp\left[-\gamma(\rho_i - \pi_j)_+\right], \tag{119}$$

where

$$(x)_+ = \max(x, 0). \tag{120}$$

If the maternal rejection system exceeds the paternal resistance system, heterospecific success declines. If paternal pollen can overcome the maternal system, the bracket is zero and the cross succeeds in this component.

For small incompatibility effects, use the same approximation as before:

$$R^{\mathrm{UI}}_{i\leftarrow j} \approx \frac{1}{2}\gamma(\rho_i - \pi_j)_+. \tag{121}$$

If rejection capacity tracks SI status s_i, with $s_i = 1$ for SI and $s_i = 0$ for SC, then a simple approximation is

$$A^{\mathrm{UI}}_{ij} \approx \frac{1}{2}\gamma(s_i - s_j). \tag{122}$$

Conclusion

The ovule-gated model predicts that asymmetry follows ovule contrast and divergence. Unilateral incompatibility predicts that asymmetry follows SI-status contrast. These are different fingerprints.

11.1 A discriminating regression

Define

$$\Delta \log O_{ij} = \log O_j - \log O_i, \tag{123}$$

and

$$\Delta s_{ij} = s_i - s_j. \tag{124}$$

A first discriminating model is

$$A_{ij} = \beta_0 + \beta_1\left[G_{ij}\Delta \log O_{ij}\right] + \beta_2 \Delta s_{ij} + \beta_3 \Delta \log O_{ij} + \epsilon_{ij}. \tag{125}$$

Feature	Ovule-gated filter	Unilateral incompatibility
Directional driver	Maternal filter set by K	SI-associated maternal rejection
Main predictor	$G_{ij}\Delta \log O_{ij}$	Δs_{ij}
Needs genetic distance to scale	Yes	Not in the simple UI term
Strongest coefficient	$\beta_1 > 0$	$\beta_2 > 0$
Expected with same ovule number	Near zero	Possible
Expected with same SI status	Possible if ovule contrast exists	Near zero
Risk	Confounded with mating system	Confounded with ovule number

11.2 The identifiability problem

If all few-ovule species are self-incompatible and all many-ovule species are self-compatible, then

$$G_{ij}\Delta \log O_{ij} \quad \text{and} \quad \Delta s_{ij} \tag{126}$$

will be collinear. The regression cannot tell the mechanisms apart. It will reward whichever predictor best captures the shared contrast.

The solution is not a better regression alone. The solution is better contrast.

Useful contrasts include:

- few-ovule SI species crossed with few-ovule SC species;
- many-ovule SI species crossed with many-ovule SC species;
- few-ovule SC species crossed with many-ovule SC species;
- few-ovule SI species crossed with many-ovule SI species;
- anatomical data on where pollen tubes fail;
- S-locus genotype or expression data;
- pollen-load or pollen-donor diversity data.

Caution

Before fitting the asymmetry model, measure the correlation between $G_{ij}\Delta \log O_{ij}$ and Δs_{ij}. If they are strongly correlated, the analysis can still be useful, but it cannot assign causality cleanly.

12 Postzygotic isolation as the contrast class

Postzygotic isolation should be modelled separately. Let B_{ij} be the Dobzhansky–Muller incompatibility load between lineages i and j. A simple model is

$$B_{ij} \sim \text{Poisson}(\lambda_B T_{ij}^{\gamma}), \tag{127}$$

where T_{ij} is divergence time. If $\gamma = 1$, incompatibilities accumulate roughly linearly. If $\gamma > 1$, incompatibilities accelerate.

Let postzygotic isolation be

$$R_{ij}^{\text{post}} = 1 - e^{-hB_{ij}}. \tag{128}$$

This barrier is mostly symmetric in the simple model:

$$R_{ij}^{\text{post}} \approx R_{ji}^{\text{post}}. \tag{129}$$

There are exceptions, such as cytoplasmic effects, endosperm imbalance, and parent-of-origin effects. Those can be added later. They should not be built into the first contrast model.

Conclusion

The clean prediction is decoupling. Prezygotic RI should track maternal competition state and show cross direction. Postzygotic RI should track divergence more strongly and show weaker direction.

13 Empirical models

13.1 Use raw counts when possible

If raw counts exist, do not collapse them too early. A proportion loses information about the denominator. A cross with 1 seed out of 2 ovules and a cross with 100 seeds out of 200 ovules have the same proportion but not the same information.

For seed counts, use

$$Y_{i\leftarrow j}^{\text{pre}} \sim \text{BetaBinomial}\left(N_{i\leftarrow j}, p_{i\leftarrow j}, \theta\right), \tag{130}$$

where $N_{i\leftarrow j}$ is the number of ovules exposed in that cross and $Y_{i\leftarrow j}^{\text{pre}}$ is the number of successful seeds.

A directed model is

$$\text{logit}(p_{i\leftarrow j}) = \beta_0 + \beta_K \log K_i + \beta_G G_{ij} + \beta_{KG} \log K_i G_{ij} + a_i^{(M)} + a_j^{(P)} + a_{ij}^{(pair)}. \tag{131}$$

Here $a_i^{(M)}$ is a maternal lineage effect, $a_j^{(P)}$ is a paternal lineage effect, and $a_{ij}^{(pair)}$ is a pair effect. In a phylogenetic model, the maternal and paternal effects should use phylogenetic covariance.

For reciprocal asymmetry, use

$$A_{ij} = \beta_0 + \beta_1 G_{ij}(\log O_j - \log O_i) + \beta_2 \Delta s_{ij} + \beta_4 \bar{t}_{ij} G_{ij}(\log O_j - \log O_i) + \epsilon_{ij}. \tag{132}$$

The expected signs are

$$\beta_1 > 0, \quad \beta_4 > 0 \tag{133}$$

when mating system increases genetically diverse pollen competition.

13.2 Phylogenetic structure

Species-pair data are not independent. The same species can appear in several crosses. The same clade can contribute many few-ovule species. Standard regressions will overstate evidence if they ignore this structure.

The empirical analysis should therefore compare at least three approaches:

1. a simple non-phylogenetic model, used only as a reference;
2. a model based on phylogenetically independent or low-overlap pairs;
3. a pairwise phylogenetic model that accounts for covariance among species pairs.

Recent pairwise comparative approaches and software such as `phylopairs` are designed for pairwise-defined traits, including reproductive isolation, where observations share taxa and therefore share history (Anderson et al., 2025).

Caution

Phylogenetic correction can remove signal if the signal is exactly a repeated transition in a trait state. The goal is not to erase ovule history. The goal is to ask whether the association remains after accounting for shared ancestry and uneven replication.

14 Current empirical data structure

The theory above gives a clean prediction. The current data decide how strongly we can test it. I separate three questions.

1. Do we have the reciprocal cross directions needed to calculate asymmetry?
2. Do both species in the pair have ovule-number estimates?
3. Are there enough comparisons within each family to distinguish a general pattern from a family-specific one?

The answer is encouraging, but it is not unlimited. The reciprocal-cross data are better than the strict PI/K2P subset. This means the asymmetric test should be the main empirical axis, with the phylogenetically independent K2P subset used as a conservative sensitivity analysis.

Conclusion

The best immediate test is not the most restrictive complete-case test. The best immediate test is the reciprocal-cross asymmetry test, with family, genus, species reuse, measure type, ovule confidence, and phylogeny handled explicitly.

14.1 Files and variables

The current data have a useful separation of roles. The cross table carries the pairwise reproductive outcomes. The lineage table carries ovule number and life-history predictors. The ovule-provenance table records how confident we should be in each ovule estimate. Genetic distances and the PI tree supply conservative divergence and phylogenetic structure.

Data object	Size	Role in the analysis
Table 1: cross outcomes	578 rows; 560 retained	Species-pair table with family, genus, species names, duplicate/removal flags, PI name, reciprocal intraspecific values, reciprocal interspecific values, hybrid proportions, RI, and measure type.
Table 2: curated lineage traits	408 species	One working ovule-number estimate per species, plus confidence, life history, and mating system. It contains 311 numeric ovule estimates and 97 missing values.
Table 2b: ovule provenance	5496 rows	Species-, genus-, and family-level ovule-number evidence. It is useful for curation, checking suspicious values, and targeted filling of missing data.
Genetic distances	141 rows; 133 unique pairs	K2P distance for PI-style pairs. There are 118 nonmissing K2P values and 23 missing values; eight rows are duplicate pair entries.
PI tree	134 tip instances; 133 unique labels	Tree for the PI labels. All retained PI labels match, but the tree has a duplicated `Corymbia` tip label that should be resolved before phylogenetic covariance modelling.

Table 1: Current data objects and their practical role in the ovule-number analysis.

Caution

The Table 1 column description appears to repeat the same maternal direction for both interspecific columns. In the audit, I treat the column names as authoritative: `interspecific_value_sp1_sp2` is species 1 mother by species 2 father, and `interspecific_value_sp2_sp1` is species 2 mother by species 1 father. This should be confirmed against the source spreadsheet before final modelling.

15 Exact asymmetric test for the current reciprocal crosses

For a retained species pair, define the four cross outcomes as

$$C_{1\leftarrow 1} = \texttt{intraspecific\_value\_sp1}, \tag{134}$$
$$C_{2\leftarrow 2} = \texttt{intraspecific\_value\_sp2}, \tag{135}$$
$$C_{1\leftarrow 2} = \texttt{interspecific\_value\_sp1\_sp2}, \tag{136}$$
$$C_{2\leftarrow 1} = \texttt{interspecific\_value\_sp2\_sp1}. \tag{137}$$

The subscripts state the direction. The symbol before the arrow is the maternal species. The symbol after the arrow is the paternal species.

The exact Sobel–Chen directional RI for species 1 as mother is

$$R^{SC}_{1\leftarrow 2} = 1 - \frac{2C_{1\leftarrow 2}}{C_{1\leftarrow 2} + C_{1\leftarrow 1}}. \tag{138}$$

We can simplify this expression:

$$R^{SC}_{1\leftarrow 2} = \frac{C_{1\leftarrow 2} + C_{1\leftarrow 1}}{C_{1\leftarrow 2} + C_{1\leftarrow 1}} - \frac{2C_{1\leftarrow 2}}{C_{1\leftarrow 2} + C_{1\leftarrow 1}} \tag{139}$$
$$= \frac{C_{1\leftarrow 1} - C_{1\leftarrow 2}}{C_{1\leftarrow 1} + C_{1\leftarrow 2}}. \tag{140}$$

The reciprocal direction is

$$R^{SC}_{2\leftarrow 1} = \frac{C_{2\leftarrow 2} - C_{2\leftarrow 1}}{C_{2\leftarrow 2} + C_{2\leftarrow 1}}. \tag{141}$$

The empirical asymmetric response is therefore

$$A^{SC}_{12} = R^{SC}_{1\leftarrow 2} - R^{SC}_{2\leftarrow 1}. \tag{142}$$

Rule used

The exact index has three useful checks. If heterospecific success equals conspecific success, then $R^{SC} = 0$. If heterospecific success is zero, then $R^{SC} = 1$. If heterospecific success exceeds conspecific success, then $R^{SC} < 0$. Negative RI is not an error; it means the heterospecific cross did better than the maternal conspecific baseline for that measure.

For the ovule predictor, use a signed log contrast. With species 1 as the first direction,

$$\Delta \log O_{12} = \log_{10}(O_2) - \log_{10}(O_1). \tag{143}$$

This sign matches the theoretical approximation in (113). If species 1 has fewer ovules than species 2, then $\Delta \log O_{12} > 0$. Under the model, species 1 should have the stricter maternal filter, so A^{SC}_{12} should be positive, all else equal.

The first asymmetric regression is

$$A^{SC}_{12} = \beta_0 + \beta_1 G_{12} \left[\log_{10}(O_2) - \log_{10}(O_1)\right] + \beta_2 \Delta s_{12} + \gamma_{m[12]} + \epsilon_{12}, \tag{144}$$

where $\gamma_{m[12]}$ is a measurement-type effect. The key prediction remains

$$\beta_1 > 0. \tag{145}$$

Caution

The simple diagnostic file from the first audit used $1 - C_{\text{inter}}/C_{\text{intra}}$ for directional RI. That is useful for quick checking, but the updated analysis should use (140)–(142). The exact index keeps the response on a symmetric scale and matches the theory above.

16 How many reciprocal comparisons are available?

The reciprocal-cross structure is strong enough for a real asymmetric test. The sample size decreases as we impose stricter biological and statistical requirements, but the drop is not catastrophic until we also require PI status and nonmissing K2P distance.

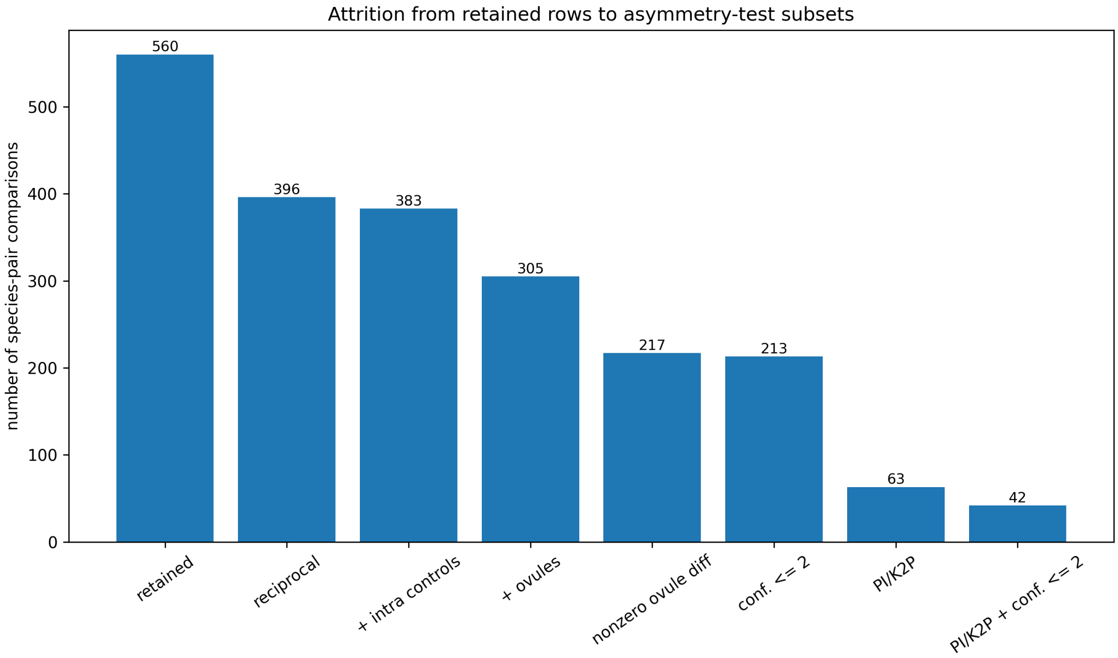

Figure 1: Attrition from all retained pair rows to the reciprocal-cross subsets needed for asymmetric tests. The main asymmetric dataset has 305 species-pair comparisons across 41 families. The strict PI/K2P version has 63 comparisons across 35 families.

Dataset condition	Pair comparisons	Families
Retained pair rows	560	50
Both reciprocal interspecific directions	396	46
Directional RI possible: reciprocal crosses plus both intraspecific controls	383	45
Asymmetry model-ready: directional RI plus both ovule numbers	305	41
Asymmetry model-ready with nonzero ovule difference	217	27
High-confidence ovules only: confidence ≤ 2 for both species	213	31
Strict PI/K2P asymmetry subset	63	35
Strict PI/K2P plus high-confidence ovules	42	27

Table 2: Data attrition for the reciprocal-cross asymmetric test.

Conclusion

The asymmetric reciprocal-cross test has 305 usable pair comparisons across 41 families. The high-confidence version has 213 comparisons across 31 families. The strict PI/K2P subset is useful, but it is too small to carry the whole paper by itself.

17 Family-level structure

The data are not evenly spread across families. This is not a nuisance detail. It tells us how to fit the model. Some families carry many reciprocal comparisons and real ovule contrasts. Other families have reciprocal crosses but no ovule contrast. Those latter families can help estimate background asymmetry and measurement noise, but they cannot estimate the ovule-number slope within family.

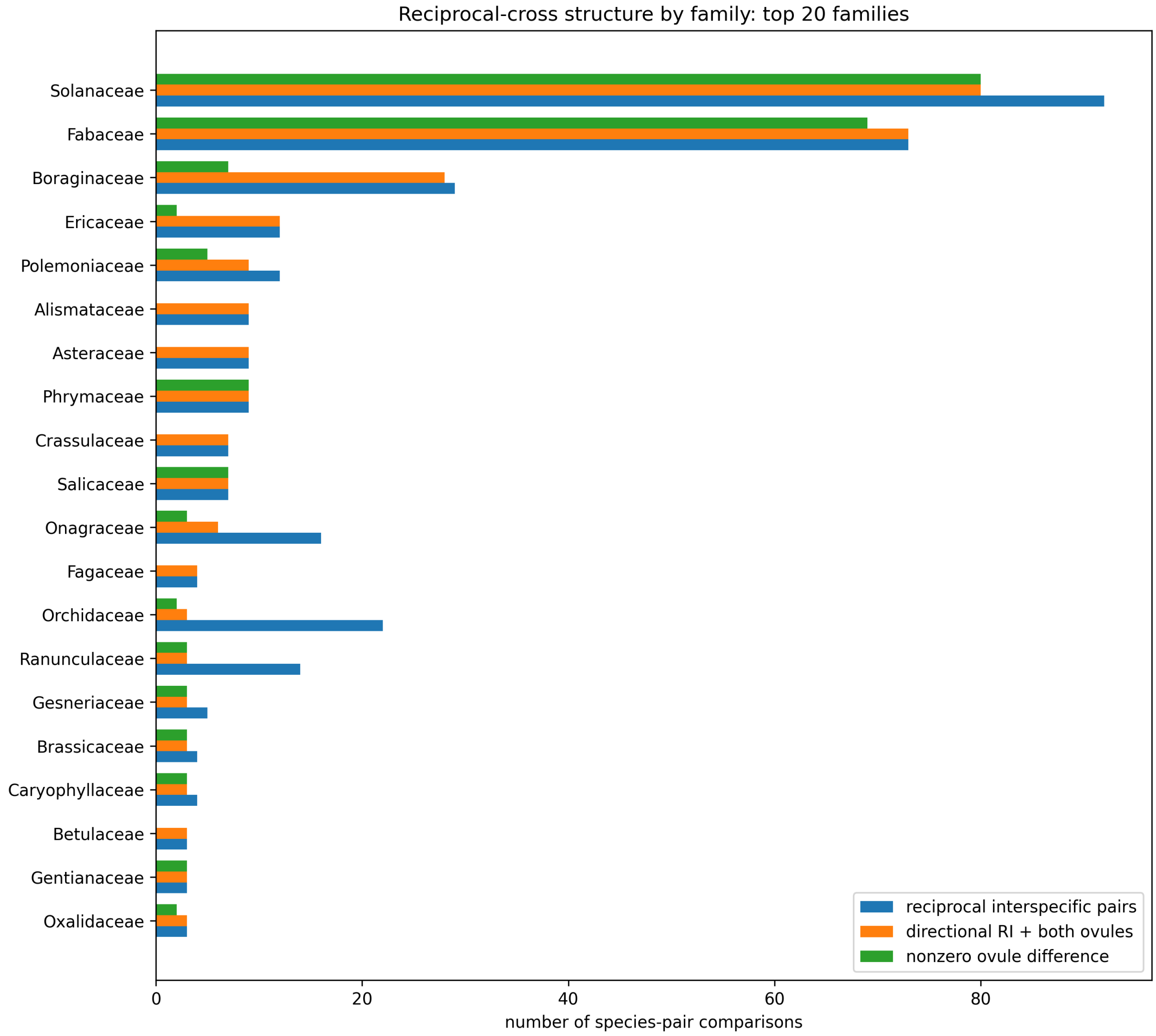

Figure 2: Family-level reciprocal-cross structure for the 20 largest families. The plotted bars count reciprocal interspecific pairs, pairs with directional RI and both ovule estimates, and pairs with nonzero ovule-number differences.

Family	Reciprocal	Directional RI	+ ovules	Nonzero ΔO	Conf. ≤ 2	PI/K2P
Solanaceae	92	87	80	80	31	7
Fabaceae	73	73	73	69	72	8
Boraginaceae	29	28	28	7	28	0
Ericaceae	12	12	12	2	1	3
Polemoniaceae	12	11	9	5	9	2
Alismataceae	9	9	9	0	9	1
Asteraceae	9	9	9	0	9	2
Phrymaceae	9	9	9	9	1	2
Crassulaceae	7	7	7	0	0	0
Salicaceae	7	7	7	7	7	2
Onagraceae	16	16	6	3	6	1
Fagaceae	4	4	4	0	4	1
Orchidaceae	22	22	3	2	2	3
Ranunculaceae	14	14	3	3	3	1
Gesneriaceae	5	5	3	3	0	2
Brassicaceae	4	4	3	3	3	3
Caryophyllaceae	4	4	3	3	3	1
Betulaceae	3	3	3	0	3	1
Gentianaceae	3	3	3	3	3	1
Oxalidaceae	3	3	3	2	3	1

Table 3: Family-level coverage for the asymmetric reciprocal-cross test. "+ ovules" means directional RI plus ovule numbers for both species. "Nonzero ΔO" means the pair has an ovule-number contrast on the log scale.

Solanaceae and Fabaceae supply 153 of the 305 asymmetry-ready comparisons. That gives power, but it also creates leverage. A pooled model could become a Solanaceae–Fabaceae model in disguise. This is why family-level analysis is not optional.

There are 19 families with at least two asymmetry-ready comparisons and at least two distinct signed ovule contrasts. There are 16 families with at least three such comparisons, and eight families with at least five. Under the confidence ≤ 2 restriction, those numbers fall to 13, 11, and six families.

Minimum family requirement	All numeric ovules	Confidence ≤ 2
At least 2 pairs and at least 2 ovule contrasts	19 families	13 families
At least 3 pairs and at least 2 ovule contrasts	16 families	11 families
At least 5 pairs and at least 2 ovule contrasts	8 families	6 families

Table 4: How many families can estimate a within-family ovule-asymmetry slope under different minimum requirements.

Conclusion

Do not fit a separate fixed slope for every family. Too many families have few comparisons. The right compromise is partial pooling: estimate a global ovule-asymmetry slope while allowing family slopes to vary around it.

17.1 A partial-pooling model for families

A practical family model is

$$A_{ij}^{SC} = \alpha + a_{f[ij]} + \left(\beta + b_{f[ij]}\right) G_{ij} \left[\log_{10}(O_j) - \log_{10}(O_i)\right] + \gamma_{m[ij]} + u_{s_1[ij]} + u_{s_2[ij]} + \epsilon_{ij}. \quad (146)$$

Here $f[ij]$ is family, $m[ij]$ is measurement type, and u_{s_1} and u_{s_2} are crossed species effects. The random family effects are

$$a_f \sim N(0, \sigma_a^2), \tag{147}$$
$$b_f \sim N(0, \sigma_b^2). \tag{148}$$

This model asks whether the average slope β is positive, while allowing families to differ.

If K2P is missing for many reciprocal pairs, fit the model twice. First fit the broader reciprocal model without G_{ij},

$$A_{ij}^{SC} = \alpha + a_{f[ij]} + \left(\beta + b_{f[ij]}\right)[\log_{10}(O_j) - \log_{10}(O_i)] + \gamma_{m[ij]} + u_{s_1[ij]} + u_{s_2[ij]} + \epsilon_{ij}. \tag{149}$$

Then fit the strict PI/K2P subset with the interaction term in (146). Agreement in sign across the broad and strict analyses is more informative than any one model alone.

Caution

Species reuse matters. Fabaceae has many *Leucaena* comparisons, and Boraginaceae is dominated by *Hydrophyllum*. Pairwise observations that share species are correlated. Crossed species effects, phylogenetic covariance, or pairwise comparative methods are needed to avoid overconfident intervals.

18 What the current data can and cannot claim

18.1 Strengths

The current datasets have five strong features. First, they contain reciprocal interspecific columns, which are essential for testing a maternal mechanism. Second, they separate cross outcomes from lineage traits, so ovule-number uncertainty can be handled directly. Third, ovule-number confidence categories exist, which makes sensitivity analysis straightforward. Fourth, there is a PI structure and a matching tree for a conservative check. Fifth, the data cover many families, so the result does not have to be presented as a single-clade pattern.

18.2 Weaknesses

The main weakness is that the strictest complete-case analysis is much smaller than the broad reciprocal analysis. Requiring reciprocal crosses, both ovule numbers, PI status, and K2P distance leaves 63 asymmetric comparisons. This subset is useful, but it cannot support many interaction terms.

The second weakness is family imbalance. Solanaceae and Fabaceae dominate the model-ready reciprocal data. They should be treated as informative families, not as independent evidence repeated 153 times.

The third weakness is that some families have reciprocal crosses but no ovule contrast. Alismataceae, Asteraceae, Crassulaceae, Fagaceae, and Betulaceae fall into this category in the current audit. These families can estimate background asymmetry but not an ovule-number slope.

The fourth weakness is measurement heterogeneity. Fruit set, seed set, seed number per fruit, and pollen competition do not share the same denominator. A measurement-type effect is mandatory. Stratified sensitivity analyses are also worth doing.

The fifth weakness is that raw denominators are mostly absent. The model is about ovules, pollen, and successful seed production, but the current table contains summary values. That means a beta-binomial or binomial count model is a future target rather than the first analysis.

Conclusion

The current data can test whether reciprocal asymmetry is aligned with signed ovule-number contrast. They cannot yet prove the full causal chain from field pollen load to pollen competition to maternal filtering. That stronger test will need pollen dose, pollen-tube, ovule-exposure, and seed-count denominators.

19 Suggested tools for the first analysis

Use three linked analyses rather than one large model.

Analysis 1: broad reciprocal asymmetry. Use all 305 asymmetry-ready pairs. Response: A_{ij}^{SC}. Predictor: signed log ovule contrast. Include family partial pooling, measure type, ovule confidence, and crossed species effects. This is the power analysis.

Analysis 2: high-confidence ovules. Repeat Analysis 1 with confidence ≤ 2 for both species. This tests whether the result depends on weak ovule estimates.

Analysis 3: strict PI/K2P. Use the 63 strict comparisons with K2P distance. Fit the theoretical interaction $G_{ij}[\log_{10}(O_j) - \log_{10}(O_i)]$. This is the conservative divergence-scaled test.

In R, the practical route is `tidyverse` for reshaping, `ape` for tree checks, `phylolm` or `nlme` for simpler phylogenetic models, and `brms` or `MCMCglmm` for crossed species effects and phylogenetic covariance. For pairwise comparative structure, `phylopairs` is worth testing alongside the PI subset. In Python, `pandas` is enough for the audit, while `bambi`, `pymc`, or `statsmodels` can fit first-pass models.

Caution

The first model should be boring. If the sign of the ovule-asymmetry effect is not stable in the boring models, a complex phylogenetic Bayesian model will not rescue the claim. Complexity should protect against false certainty, not create the signal.

20 Immediate data-cleaning checklist

Before fitting the models, make these changes.

1. Add an explicit Boolean `is_PI` column to Table 1. Keep `PI_name` as the tree label.
2. Confirm the reciprocal direction of `interspecific_value_sp2_sp1` against the source spreadsheet.

3. Collapse duplicate genetic-distance rows and keep one K2P value per unordered species pair.
4. Resolve the duplicated `Corymbia` tree tip before constructing phylogenetic covariance matrices.
5. Recompute directional RI using the exact Sobel–Chen formula in (140)–(141).
6. Mark whether each pair has a nonzero signed ovule contrast. Families with no contrast should not be used to estimate within-family ovule slopes.
7. Keep measurement type in every model and run stratified checks for seed set and fruit set.

Conclusion

The next file we should build is a single analysis table with one row per reciprocal pair and the following fields: family, genus, species 1, species 2, measure, exact directional RI in both directions, exact asymmetry, both ovule numbers, both ovule confidence scores, signed log ovule contrast, mating-system contrast, life-history contrast, K2P distance, PI flag, and tree labels.

21 Missing data

21.1 Missing pollen-load data

Direct pollen-load data are unavailable for the present analysis. This is a limitation, but it does not stop the model. Treat pollen load as a latent quantity.

The simplest approach is sensitivity analysis:

$$\psi \in \{0.001, 0.003, 0.01, 0.02, 0.05, 0.10\}, \tag{150}$$

and

$$\alpha \in \{0.25, 0.50, 0.75, 1.00, 1.25\}. \tag{151}$$

For each combination, compute

$$K_i = \psi c O_i^{\alpha-1}. \tag{152}$$

Then ask whether the same qualitative result appears across a broad region of the grid.

A Bayesian latent-variable model is

$$\log K_i = \mu_K + (\alpha - 1)\log O_i + \eta_t \operatorname{logit}(t_i) + u_i + \epsilon_i, \tag{153}$$

where u_i is a phylogenetic effect and ϵ_i is lineage-specific noise. The term $\eta_t \operatorname{logit}(t_i)$ allows outcrossing rate to modify effective pollen competition.

Conclusion

The missing pollen-load data become a calibration problem. The model is more credible if the sign of the prediction depends on $\alpha < 1$, not on one tuned value of ψ.

21.2 Missing reciprocal crosses

Reciprocal crosses are central because the strongest prediction is directional. If only one cross direction is available, the data can test whether few-ovule lineages have stronger RI on average. They cannot test the maternal-filter prediction cleanly.

21.3 Missing mating-system data

If outcrossing rate is missing, code broad categories with uncertainty:

$$t_i \in \{\text{selfing}, \text{mixed}, \text{outcrossing}\}. \tag{154}$$

Then run the analysis under alternative codings. For example, assign approximate values

$$t_i = 0.05,\ 0.50,\ 0.95 \tag{155}$$

for selfing, mixed, and outcrossing. Then repeat with wider values. The goal is to test whether the ovule signal depends on implausible mating-system assignments.

21.4 Worst-to-best data scenarios

Scenario	Available data	Claim allowed	Main risk
Worst publishable	Ovule number, summary RI, rough phylogeny	Theory plus suggestive comparative pattern	Cannot estimate K or direction well
Minimal directional	Ovule number, genetic distance, some reciprocal RI	Test sign of asymmetry with ovule contrast	Pollen load and mating system inferred
Solid comparative	Raw cross counts, ovule denominators, reciprocals, postzygotic data, phylogeny	Test prezygotic-postzygotic decoupling	K still partly latent
Strong mechanistic	Adds pollen tubes, pollen production, mating system, SI status	Locate barrier and separate UI	Field pollen load still limited
Best empirical	Adds natural pollen load, donor diversity, timing, manipulative pollen-dose experiments	Test the causal chain directly	Hardest to collect

22 Simulation protocol

The simulation should answer two questions. First, does the mathematical mechanism behave as expected? Second, what data structures can recover the mechanism?

22.1 Layer 1: ecological competition without RI

Simulate lineages with ovule number O_i. For each lineage, simulate effective pollen load using

$$P_{if} \sim \text{NegBinomial}(\mu_i, \theta_P), \tag{156}$$

where

$$\mu_i = \psi c O_i^{\alpha}. \tag{157}$$

Compute

$$K_{if} = P_{if}/O_i \tag{158}$$

and

$$S_z(K_{if}) \tag{159}$$

using (37).

Record mean K, Ω, and S_z for each lineage. This layer tests only the competition engine.

22.2 Layer 2: evolving maternal filters

Let filter strictness evolve toward

$$F_i^* = F_{\max} \frac{K_i}{K_i + K_{50}}, \tag{160}$$

where $F_{\max}$ is the maximum filter and K_{50} is the value of K at which the filter reaches half of $F_{\max}$.

This simulation function is simpler than the fitness model above. It keeps the key property:

$$\frac{dF^*}{dK} > 0. \tag{161}$$

Simulate mismatch as

$$D_{ij} = \delta_D G_{ij} + D_{ij}^{\text{arch}}, \tag{162}$$

where D_{ij}^{arch} depends on genetic architecture.

22.3 Layer 3: genetic architectures

Use three architectures.

Independent architecture. Pollen-performance loci and compatibility loci are unlinked. Selection changes pollen performance, but RI evolves slowly unless mismatch also accumulates.

Linked architecture. Pollen-performance loci are linked to compatibility loci. Hitchhiking strength can be approximated by

$$H(r, s) = \frac{s}{s + r}, \tag{163}$$

where s is the selected effect and r is recombination. Low recombination strengthens passenger RI.

Pleiotropic architecture. The same allele affects pollen performance and compatibility. Let the association between performance effects and mismatch effects be $\rho_{a,d}$. Passenger RI should evolve fastest when $\rho_{a,d}$ is high.

Conclusion

The expected order is pleiotropy > linkage > independence for speed and strength of passenger RI.

22.4 Layer 4: phylogenetic embedding

Simulate trees with different sizes:

$$n \in \{30, 60, 120, 240\}. \tag{164}$$

Simulate ovule number under three models:

1. Brownian motion on $\log O$;
2. an Ornstein–Uhlenbeck model on $\log O$;
3. a discrete transition model between few-ovule and many-ovule regimes.

The third model matters most for the comparative question because it counts independent transitions.

22.5 Layer 5: crossing data

For each pair, compute

$$R_{i \leftarrow j} = \tanh\left(\frac{F_i D_{ij}}{2}\right), \tag{165}$$

$$R_{j \leftarrow i} = \tanh\left(\frac{F_j D_{ij}}{2}\right), \tag{166}$$

and

$$A_{ij} = R_{i \leftarrow j} - R_{j \leftarrow i}. \tag{167}$$

Simulate observed seed counts:

$$Y_{i \leftarrow j} \sim \text{BetaBinomial}(N_{i \leftarrow j}, 1 - R_{i \leftarrow j}, \theta). \tag{168}$$

Simulate postzygotic RI from (127) and (128).

22.6 Layer 6: damage the data

Create data structures from best to worst:

1. all reciprocal crosses, raw counts, pollen load, SI status, mating system;
2. reciprocal crosses and raw counts, but no pollen load;
3. reciprocal crosses for a sparse set of pairs;
4. one-direction crosses only;
5. summary proportions only;
6. family-level states only.

For each structure, fit (132) or (125). Record bias, power, false-positive rate, and sign recovery.

22.7 Core simulation outputs

Record

$$K_i, \quad K_{\mathrm{sex},i}, \quad \Omega_i, \quad S_z(K_i), \quad F_i, \quad D_{ij}, \quad R_{i\leftarrow j}, \quad A_{ij}, \quad R_{ij}^{\mathrm{post}}. \tag{169}$$

Also record genomic summaries:

number of sweeps,
mean sweep strength,
fraction of sweeps affecting compatibility,
fraction of hitchhiked compatibility alleles.

Conclusion

The simulation should be judged by recovery, not by appearance. The main recovery target is a positive coefficient for $G_{ij}(\log O_j - \log O_i)$ after accounting for SI status and mating system.

23 Collaborator checklist

Before analysing the real data, answer these questions.

1. Do we have reciprocal crosses for the pairs that matter most? The current answer is yes for 396 retained pairs, and 305 also have both ovule estimates.
2. Which families estimate the ovule slope rather than only background asymmetry? The current answer is about 19 families under a lenient rule and eight under a five-pair rule.
3. Are the few-ovule taxa also mostly self-incompatible?
4. Are the many-ovule taxa also mostly self-compatible?
5. Do we have raw counts or only summary RI values?
6. Do we know whether the barrier acts at stigma, style, ovary, ovule, seed set, or F1 fertility?
7. Do we know mating system well enough to separate selfers, mixed-maters, and outcrossers?
8. Are few-ovule states replicated across the phylogeny, or concentrated in one clade?
9. Does asymmetry occur at intermediate divergence, where the model says it should be most visible?
10. Does the sign of the exact Sobel–Chen asymmetry agree with the sign of the simpler diagnostic ratio asymmetry?

Conclusion

The strongest test is not a correlation between ovule number and RI. The strongest test is directional: reciprocal asymmetry should follow maternal competition state, after controlling for divergence, SI status, mating system, and phylogeny.

24 Summary of the model

The model can be written as one causal chain:

$$O_i \to K_i = \frac{P_{\mathrm{eff},i}}{O_i} \to S_z(K_i) \to F_i^*(K_i) \to R_{i\leftarrow j}. \tag{170}$$

Under sublinear pollen-load scaling,

$$P_{\mathrm{eff},i} = \psi c O_i^{\alpha}, \quad \alpha < 1, \tag{171}$$

so

$$K_i = \psi c O_i^{\alpha-1}. \tag{172}$$

Few-ovule lineages then have higher K. Higher K strengthens pollen competition and makes maternal filtering less costly. When pollen-pistil mismatch exists between lineages, stricter maternal filtering creates directional prezygotic RI.

The central empirical approximation is

$$A_{ij} \approx \beta_1 G_{ij}(\log O_j - \log O_i), \quad \beta_1 > 0. \tag{173}$$

This prediction should be strongest in outcrossing or mixed-mating lineages with genetically diverse pollen loads. It should weaken in predominant selfers.

Conclusion

The claim is narrow enough to test. Reproductive isolation is not treated as an inevitable clock. Prezygotic RI can be regenerated when reproductive ecology places lineages in a high pollen-competition state.

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